

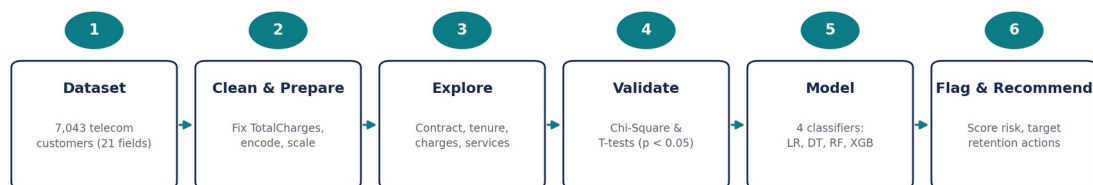
Telecom Customer Churn Prediction Using Statistical Analysis and Machine Learning:

A Data-Driven Framework for Identifying At-Risk Customers

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WORKFLOW SUMMARY: Dataset → Cleaning & Preparation → Exploratory Analysis → Statistical Validation → Model Training & Comparison → Risk Flagging & Recommendations

Figure 1: Graphical overview of the proposed churn analysis and prediction framework, from raw customer data to flagged, actionable risk insights.

The graphical abstract illustrates the overall workflow of the proposed churn analysis framework. The process begins with a raw export of 7,043 telecom customer records, followed by data cleaning and preparation to fix data types, encode categorical fields, and scale numerical features. Exploratory analysis is then used to characterize churn across contract type, tenure, charges, and service usage. Three statistical hypothesis tests validate that the observed relationships are significant rather than coincidental. Four machine learning classifiers are trained and compared, and the best-performing model is used to identify the leading churn drivers and flag at-risk customers for targeted retention action.

Highlights

Telecom Customer Churn Prediction Using Statistical Analysis and Machine Learning

Narmatha Muralidharan

- Analyzed 7,043 real telecom customer records to identify the key behavioral and account-level drivers of customer churn.
- Cleaned and prepared the dataset, encoding categorical variables and scaling numerical features for machine learning.
- Validated exploratory findings using three statistical hypothesis tests (two Chi-Square tests and one T-test), all statistically significant at $p < 0.05$.
- Built and compared four classification models — Logistic Regression, Decision Tree, Random Forest, and XGBoost — evaluated on accuracy, precision, recall, F1-score, and AUC-ROC.
- Identified Total Charges, Monthly Charges, Tenure, Contract Type, and Payment Method as the strongest predictors of churn using feature importance analysis.
- Translated the findings into concrete, actionable business recommendations for telecom customer retention.

Abstract

Customer churn — the loss of subscribers to a competitor or to service cancellation — is one of the most pressing revenue challenges facing telecommunications companies, since acquiring a new customer is estimated to cost five to seven times more than retaining an existing one. This project develops a complete data analytics pipeline to understand and predict customer churn using a real, publicly available dataset of 7,043 telecom customers.

The proposed framework includes data cleaning and preprocessing, exploratory data analysis (EDA) across demographic, account, and service-usage features, statistical hypothesis testing to validate observed patterns, and the training and comparison of four supervised machine learning classifiers: Logistic Regression, Decision Tree, Random Forest, and XGBoost. Model performance was evaluated using accuracy, precision, recall, F1-score, and AUC-ROC on a held-out 20% test set.

The results show an overall churn rate of 26.5%, with month-to-month contracts, low tenure, high monthly charges, and the absence of support services all significantly associated with higher churn. Three hypothesis tests confirmed that contract type, monthly charges, and internet service type each have a statistically significant relationship with churn ($p < 0.001$ in all cases). Among the four models tested, Logistic Regression achieved the best overall performance, with 80.06% accuracy and an AUC-ROC of 0.8403, while feature importance analysis identified Total Charges, Monthly Charges, and Tenure as the leading predictors.

This project demonstrates that a combination of statistically grounded exploratory analysis and interpretable machine learning can provide telecom companies with an actionable, auditable framework for proactively identifying at-risk customers and designing targeted retention strategies.

Keywords

Customer Churn Prediction, Telecommunications, Machine Learning, Logistic Regression, XGBoost, Exploratory Data Analysis, Statistical Hypothesis Testing, Customer Retention

1. Introduction

1.1 Application Background and Practical Importance

The telecommunications industry operates in a highly saturated and competitive market, where subscribers can switch providers with relatively little friction. Customer churn — the rate at which subscribers discontinue their service — directly erodes recurring revenue and increases customer acquisition costs, since replacing a lost customer is widely estimated to cost five to seven times more than retaining an existing one. As telecom markets mature, retention has become a central lever for sustaining profitability.

Traditional churn management has often relied on reactive measures, such as win-back offers issued only after a customer has already cancelled, or broad, untargeted retention campaigns applied uniformly across the customer base. These approaches are inefficient, as they either act too late or waste resources on customers who were never at meaningful risk of leaving. What telecom companies need instead is a proactive, data-driven approach that can identify which customers are likely to churn before they do, and why.

This project addresses that need by developing a churn analysis and prediction framework using a real, structured telecom customer dataset. The framework combines exploratory data analysis, statistical hypothesis testing, and supervised machine learning to both explain the drivers of churn and predict which individual customers are most at risk.

1.2 Data Analytics and Predictive Modeling Context

Customer churn prediction is a well-established binary classification problem in applied data analytics: given a set of customer attributes, the objective is to predict whether a customer will churn (Yes) or remain (No). Unlike problems where ground truth is scarce or unavailable, telecom churn datasets typically include a confirmed historical outcome label, making supervised machine learning directly applicable.

Because the target class is imbalanced — in this dataset, only 26.5% of customers churned — model evaluation cannot rely on accuracy alone, since a model that always predicts “No churn” would already be correct roughly three-quarters of the time while being practically useless. This project therefore evaluates every model using a fuller set of metrics — precision, recall, F1-score, and AUC-ROC — that are more informative under class imbalance.

1.3 Machine Learning for the Selected Problem

This project compares four classification algorithms that represent increasing levels of model complexity and different underlying assumptions:

- Logistic Regression — a linear, highly interpretable baseline model that estimates the probability of churn as a function of customer attributes.
- Decision Tree — a rule-based model that recursively splits customers on feature thresholds, offering an intuitive, if-then view of churn risk.
- Random Forest — an ensemble of many decision trees that reduces overfitting and typically improves generalization over a single tree.
- XGBoost — a gradient-boosted tree ensemble that iteratively corrects prior errors, generally considered a strong performer on structured, tabular data.

Comparing these four models allows the project to assess whether the added complexity of ensemble methods (Random Forest, XGBoost) provides a meaningful performance benefit over a simple, interpretable linear baseline (Logistic Regression) for this particular dataset.

1.4 Model Evaluation and Interpretation

Each model was trained on 80% of the cleaned dataset and evaluated on the remaining 20%, held out and stratified by churn status to preserve the original class balance in both splits. Evaluation was performed using accuracy, precision, recall, F1-score, and AUC-ROC, and the best-performing model was additionally examined using a confusion matrix and ROC curve to understand its class-level prediction behavior. Feature importance was extracted from the trained models to identify which customer attributes most strongly influence churn predictions.

1.5 Research Gap

Much of the publicly available analysis of telecom churn datasets focuses narrowly on either descriptive exploratory statistics or a single predictive model, without connecting the two. Descriptive analyses alone do not produce an actionable predictive tool, while purely predictive analyses often skip the statistical validation step that would confirm whether an observed pattern is a genuine, significant relationship or simply noise in the data.

This project addresses that gap by combining exploratory data analysis, formal statistical hypothesis testing, and a comparative evaluation of multiple machine learning models within a single, reproducible pipeline, ensuring that every reported churn driver is both descriptively visible and statistically validated before being used to justify a business recommendation.

1.6 Problem Statement

Telecom companies need an efficient, data-driven method to identify customers who are likely to churn so that retention efforts can be proactively targeted rather than applied broadly or only after a customer has already left. This project addresses the problem of classifying telecom customers as likely to churn or likely to remain, using structured account, demographic, and service-usage data, and of identifying which specific factors most strongly drive that classification.

1.7 Aim and Objectives

The aim of this study is to develop and evaluate a data-driven framework for understanding and predicting telecom customer churn, using exploratory analysis, statistical testing, and machine learning classification.

The objectives of this study are:

1. To collect and structurally analyze a real telecom customer dataset, including demographic, account, and service-usage attributes.
2. To clean the dataset and prepare it for analysis, handling missing values and encoding categorical fields.
3. To conduct exploratory data analysis characterizing churn across contract type, payment method, tenure, monthly charges, and support services.
4. To statistically validate the most important observed relationships using hypothesis testing.
5. To build and compare four machine learning classification models for predicting customer churn.

6. To identify the leading predictors of churn using feature importance analysis and translate the findings into business recommendations.

1.8 Research Questions

This project is guided by the following research questions.

RQ1 How accurately can machine learning classification models predict telecom customer churn using structured account, demographic, and service-usage data?

RQ2 Which customer attributes — contract type, tenure, monthly charges, or service usage — are most strongly and significantly associated with churn?

RQ3 Do the resulting model outputs and identified churn drivers provide a practical basis for targeted customer retention strategies?

1.9 Contributions and Novelty

This project presents a practical, end-to-end implementation of churn analytics on a real telecom dataset. Rather than proposing a new algorithm, its contribution lies in combining descriptive, statistical, and predictive methods into a single coherent pipeline and evaluating the results comprehensively. The main contributions of this project are as follows:

- Development of a complete churn analysis pipeline covering data cleaning, exploratory analysis, statistical testing, model building, and business interpretation.
- Statistical validation of exploratory findings using Chi-Square and T-tests, confirming that observed churn patterns are significant rather than coincidental.
- Comparative evaluation of four machine learning classifiers under a consistent experimental setup.
- Feature importance analysis identifying and ranking the leading predictors of customer churn.
- Translation of statistical and model-based findings into concrete, actionable retention recommendations.

1.10 Report Organization

The remainder of this report is organized as follows. Section 2 reviews related literature on customer churn prediction in telecommunications. Section 3 presents the methodology, including dataset description, data preparation, statistical testing procedures, and model development. Section 4 describes the implementation environment and ethical considerations. Section 5 presents the experimental results. Section 6 discusses the findings, limitations, and future improvements. Section 7 concludes the report.

2. Literature Review

2.1 Customer Churn in the Telecommunications Industry

Customer churn has long been recognized as one of the most damaging outcomes for subscription-based businesses, and telecommunications has served as one of the primary domains for churn prediction research due to the availability of detailed, structured customer data. Verbeke, Dejaeger, Martens, Hur, and Baesens (2012) provide a comprehensive review of churn prediction techniques in the telecommunications sector, emphasizing that both predictive accuracy and model interpretability are important, since retention teams need to understand why a customer is at risk, not just that they are.

2.2 Statistical and Machine Learning Approaches to Churn Prediction

A range of statistical and machine learning techniques have been applied to churn prediction, from traditional logistic regression to ensemble tree-based methods. Lemmens and Croux (2006) demonstrated that bagging and boosting techniques applied to classification trees can meaningfully improve churn prediction performance over single-tree models, a finding that motivates this project's inclusion of both Random Forest (bagging) and XGBoost (boosting) alongside simpler baseline models.

2.3 Publicly Available Telecom Churn Data

The dataset used in this project follows the structure of the widely used IBM Telco Customer Churn dataset, distributed publicly via Kaggle. This dataset has become a standard benchmark for churn prediction research and coursework because it combines demographic, account, and service-usage fields with a confirmed churn outcome label, enabling both descriptive analysis and supervised model training on the same structured data.

2.4 Determinants of Customer Churn

Across the churn prediction literature, a recurring set of factors is consistently associated with higher churn risk: short contract commitment (particularly month-to-month plans), low customer tenure, high monthly billing relative to perceived value, and the absence of bundled support or security services. These same factors directly informed the exploratory analysis and hypothesis tests conducted in this project.

2.5 Model Evaluation in Imbalanced Classification

Because churn datasets are typically imbalanced — with far fewer churned customers than retained ones — evaluation practice in the literature consistently recommends reporting precision, recall, F1-score, and AUC-ROC alongside accuracy, since accuracy alone can be misleadingly high for a model that is biased toward the majority class. This project follows that established practice throughout its model evaluation (Section 3.7 and Section 5).

2.6 Comparative Literature Review Matrix

Table 1 compares representative approaches to churn analysis with the framework proposed in this project, based on dataset, model, task, and evaluation approach.

Study Focus	Dataset	Model	Task	Evaluation Approach
Telecom churn prediction review	Multiple telecom	Multiple (survey)	Churn prediction	Accuracy, interpretability

Study Focus	Dataset	Model	Task	Evaluation Approach
	datasets			
Ensemble tree methods for churn	Telecom subscriber data	Bagging / boosting trees	Churn classification	Predictive accuracy (AUC)
Open benchmark churn dataset	IBM Telco (Kaggle)	—	Data publication	Data quality documentation
This Work	Telecom customers (7,043)	Logistic Regression, Decision Tree, Random Forest, XGBoost	Churn prediction & driver analysis	Accuracy, precision, recall, F1, AUC-ROC, hypothesis testing

2.7 Synthesis of Literature and Identified Gap

The literature confirms that contract structure, tenure, billing level, and service bundling are reliable, recurring predictors of telecom churn, and that ensemble methods can improve on simple baselines, though often at the cost of interpretability. However, many applied analyses either stop at descriptive statistics or move directly to a single predictive model without formally testing whether the observed patterns are statistically significant. This project addresses that gap by combining exploratory analysis, formal hypothesis testing, and a multi-model comparative evaluation within a single, reproducible pipeline, directly motivating the methodology presented in the following chapter.

3. Methodology

3.1 Research Design

This study follows an exploratory, quantitative, and predictive research design for customer churn analysis. The objective is to both understand which customer attributes drive churn and to build an accurate, interpretable classification model for predicting it.

The study follows a complete data analytics pipeline: dataset collection and structural analysis, data cleaning and preprocessing, exploratory data analysis, statistical hypothesis testing, feature engineering, model building, and model evaluation. The entire implementation was carried out in a Google Colab notebook using Python and the Pandas, NumPy, Matplotlib, Seaborn, SciPy, Scikit-learn, and XGBoost libraries.

3.2 Dataset Description

3.2.1 Dataset Source

The dataset used in this study is the IBM Telco Customer Churn dataset, publicly distributed via Kaggle. It is freely accessible for academic use and was loaded directly into the analysis environment as a CSV file using the Pandas library, ensuring the proposed framework is fully reproducible.

3.2.2 Dataset Composition

The dataset contains 7,043 customer records described by 21 columns, spanning customer demographics, account information, subscribed services, and the confirmed churn outcome. Table 2 summarizes the feature groups used in this analysis.

Feature Group	Fields Included
Demographics	gender, SeniorCitizen, Partner, Dependents
Account information	tenure, Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges
Services	PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies
Target	Churn (Yes/No)

3.2.3 Field Definitions

The target field, Churn, records whether a customer discontinued their service (Yes) or remained (No). The Contract field records the customer's billing commitment (month-to-month, one year, or two year). The tenure field records the number of months the customer has stayed with the company. MonthlyCharges and TotalCharges record the customer's current monthly bill and cumulative billing to date, respectively. Service fields such as TechSupport and OnlineSecurity record whether the customer has each add-on service enabled. These fields form the primary inputs to both the exploratory analysis and the predictive models described in this report.

3.3 Data Preparation

3.3.1 Data Cleaning

Before analysis, the following cleaning steps were applied to the raw dataset:

7. Missing value handling. The TotalCharges field, stored as a text field, contained blank entries for a small number of customers with zero tenure (newly joined customers who had not yet been billed). These entries were converted to numeric type and the corresponding rows handled explicitly rather than silently dropped.
8. Data type conversion. TotalCharges was converted from a text/object type to a numeric (float) type to enable statistical and machine learning use.
9. Column review. Non-informative identifier fields (customerID) were excluded from modeling.
10. Duplicate check. The dataset was checked for duplicate customer records.

3.3.2 Feature Engineering

To support exploratory analysis, a derived TenureGroup field was created by binning the continuous tenure field into five interpretable bands (0–12, 13–24, 25–48, 49–60, and 61–72 months), allowing churn rate to be examined across customer lifecycle stages.

3.3.3 Encoding and Scaling

For machine learning, all categorical columns (including gender, Contract, PaymentMethod, and the service fields) were transformed into numeric form using label encoding. The three continuous numerical fields used directly by the models — tenure, MonthlyCharges, and TotalCharges — were standardized using a StandardScaler, which rescales each feature to zero mean and unit variance, ensuring that no single feature dominates model training due to differences in scale.

3.3.4 Train–Test Split

The cleaned, encoded dataset was split into training and testing subsets using an 80/20 split (5,634 training records and 1,409 testing records), with stratification on the Churn target to preserve the original 26.5% churn rate in both subsets. A fixed random seed (random_state = 42) was used throughout to ensure the split and all model results are fully reproducible.

3.4 Exploratory Data Analysis Methodology

Exploratory data analysis was conducted to characterize how churn varies across contract type, payment method, tenure, monthly and total charges, and key support services (tech support, online security), and to examine pairwise correlations between numerical features and the churn outcome using a correlation heatmap. These visualizations, presented in Section 5.2, directly motivated the choice of variables tested statistically in Section 3.5 and used as model inputs in Section 3.6.

3.5 Statistical Hypothesis Testing Methodology

To confirm that patterns observed during EDA reflect genuine, non-random relationships, three hypothesis tests were conducted at a significance threshold of $\alpha = 0.05$:

- Chi-Square test of independence (Churn vs. Contract Type) — tests whether churn and contract type are statistically independent categorical variables.
- Independent samples T-test (Monthly Charges: churned vs. retained customers) — tests whether the mean monthly charge differs significantly between the two churn groups.
- Chi-Square test of independence (Churn vs. Internet Service Type) — tests whether churn and internet service type are statistically independent categorical variables.

A result is considered statistically significant when the resulting p-value is below 0.05, indicating less than a 5% probability that the observed relationship arose by random chance alone.

3.6 Model Development

Four supervised classification models were trained on the prepared training set using consistent preprocessing:

Model	Configuration
Logistic Regression	max_iter = 1000, random_state = 42
Decision Tree	max_depth = 5, random_state = 42
Random Forest	n_estimators = 100, random_state = 42
XGBoost	n_estimators = 100, random_state = 42, eval_metric = logloss

3.7 Evaluation Framework

Each trained model was evaluated on the held-out 20% test set using five metrics: accuracy (overall proportion of correct predictions), precision (proportion of predicted churners who actually churned), recall (proportion of actual churners correctly identified), F1-score (the harmonic mean of precision and recall), and AUC-ROC (the model's ability to discriminate between churners and non-churners across all classification thresholds). Because the churn class is imbalanced, precision, recall, F1-score, and AUC-ROC were treated as equally important as accuracy when comparing models. The best-performing model was further examined using a confusion matrix, an ROC curve comparison across all four models, and a feature importance ranking.

4. Implementation Environment and Ethical Considerations

The entire implementation was carried out using Python in a Google Colab notebook environment. Data cleaning, feature engineering, and statistical testing were performed using Pandas, NumPy, and SciPy; visualization was performed using Matplotlib and Seaborn; and machine learning models were built using Scikit-learn and XGBoost. The Colab environment provided a reproducible, browser-accessible interface for the full pipeline, from data loading through model evaluation, without requiring any external software installation.

The dataset used in this study consists of anonymized, structured customer account and service-usage data with no directly identifying personal information beyond an internal customer ID, which was excluded from all analysis and modeling. The dataset is used solely for academic research and educational purposes.

The developed model is intended as a decision-support tool to help prioritize retention outreach, not as an automated system for taking action against individual customers. Predictions should be reviewed by a retention or customer-success team before being used to make consequential decisions, such as offering a discount or contacting a customer. As with any predictive model trained on historical data, care should be taken to periodically re-evaluate model performance as customer behavior and market conditions evolve over time.

5. Results

5.1 Dataset and Preprocessing Results

The raw dataset was successfully loaded and structurally analyzed, confirming 7,043 customer records across 21 columns. The TotalCharges field was successfully converted to a numeric type, and the dataset was verified to contain no unresolved missing values after cleaning. All categorical fields were successfully label-encoded, and the three continuous numerical fields (tenure, MonthlyCharges, TotalCharges) were successfully scaled. The resulting dataset was split into 5,634 training records and 1,409 testing records, with the 26.5% churn rate preserved in both subsets.

5.2 Exploratory Data Analysis Results

5.2.1 Churn Distribution

Of the 7,043 customers in the dataset, 5,174 (73.5%) were retained and 1,869 (26.5%) churned, as shown in Figure 2. This confirms that more than one in four customers left the service during the observed period — a churn rate substantial enough to warrant proactive, targeted retention efforts.

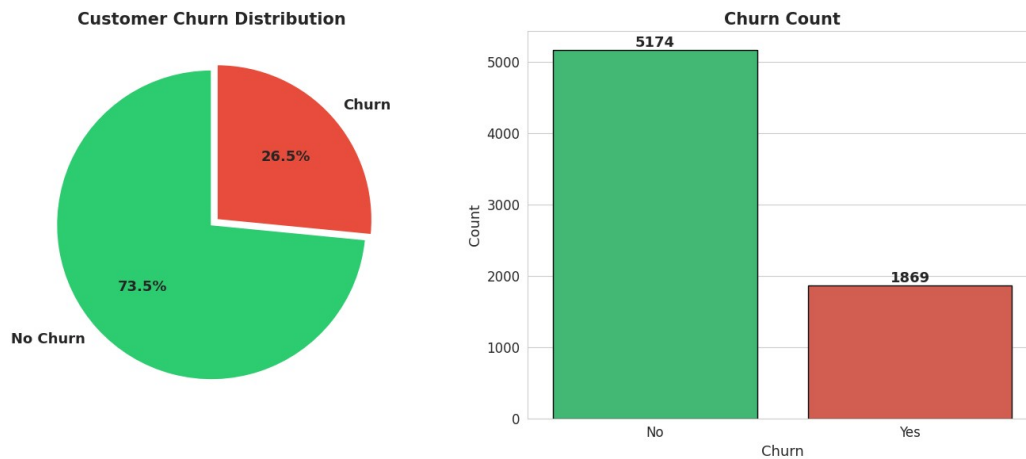


Figure 2: Overall customer churn distribution.

5.2.2 Churn by Contract Type and Payment Method

Figure 3 shows that churn is strongly concentrated among month-to-month customers, who churn at a rate of 42.7%, compared to 11.3% for one-year contracts and just 2.8% for two-year contracts. Figure 4 shows a similar pattern for payment method: customers paying by electronic check churn at the highest rate among all payment methods, at 45.3%, compared to roughly 15–19% for the other three payment methods.

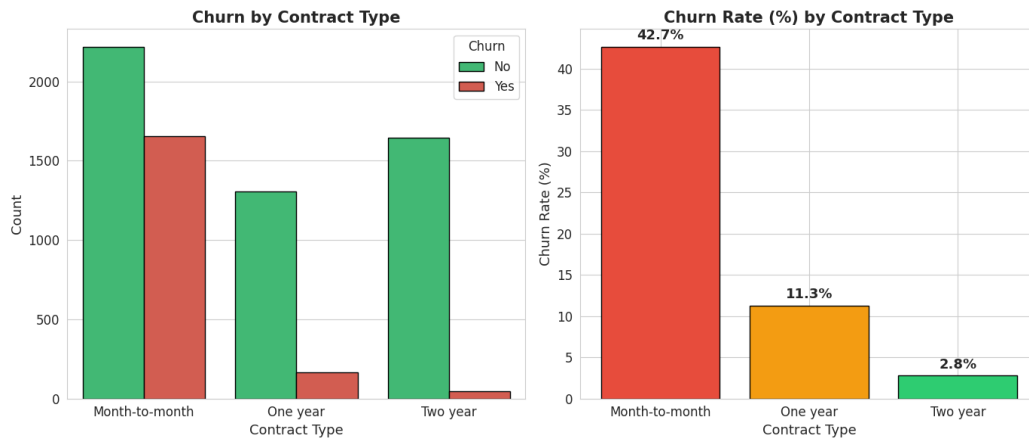


Figure 3: Churn count and churn rate by contract type.

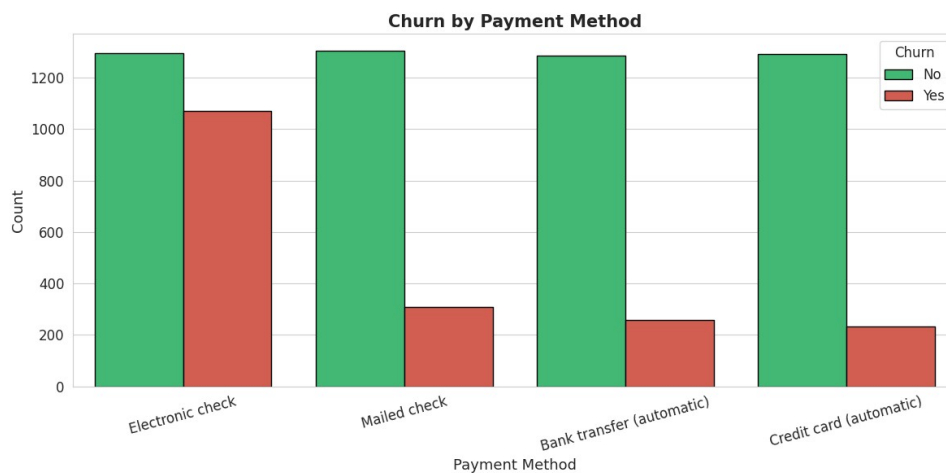


Figure 4: Churn by payment method.

5.2.3 Tenure and Monthly Charges

Figure 5 shows that churned customers have a markedly lower average tenure (18.0 months) than retained customers (37.6 months), with churn heavily concentrated among customers in their first few months of service. Figure 6 shows that churned customers also pay higher average monthly charges (\$74.44) than retained customers (\$61.27), indicating that customers who feel they are paying more relative to their tenure and perceived value are more likely to leave.

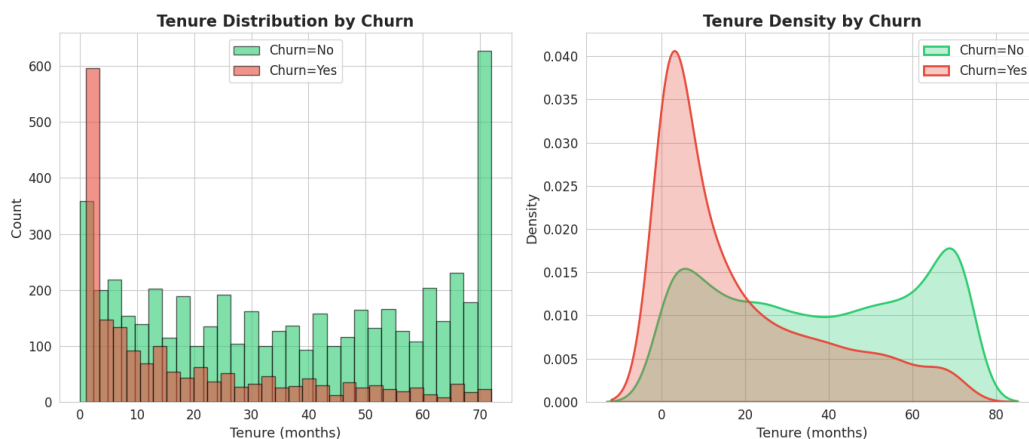


Figure 5: Tenure distribution and density by churn status.

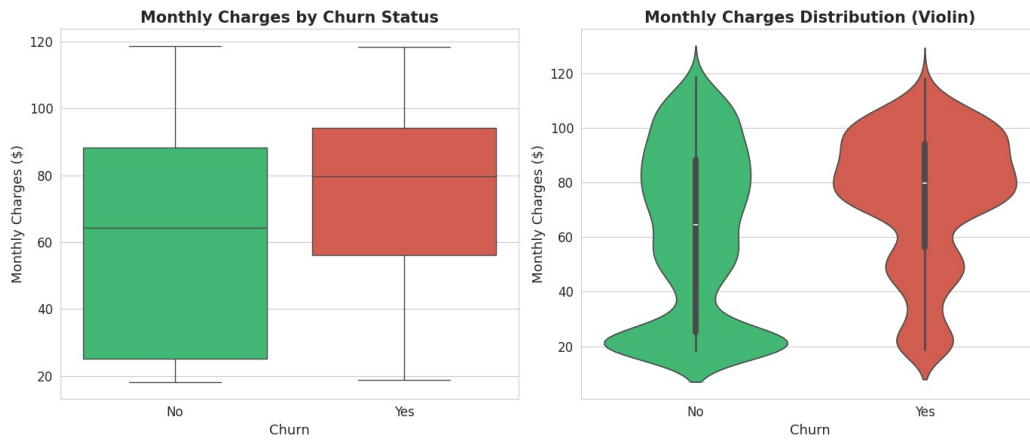


Figure 6: Monthly charges by churn status.

5.2.4 Support Services and Correlation Analysis

Figure 7 shows that customers without tech support churn at 41.6%, compared to just 15.2% for customers who have tech support enabled; a nearly identical pattern holds for online security. Figure 8 presents a correlation heatmap of all numerical and encoded features against churn, showing that Contract (-0.397), tenure (-0.352), and OnlineSecurity (-0.289) carry the strongest linear associations with the churn outcome among the examined features.

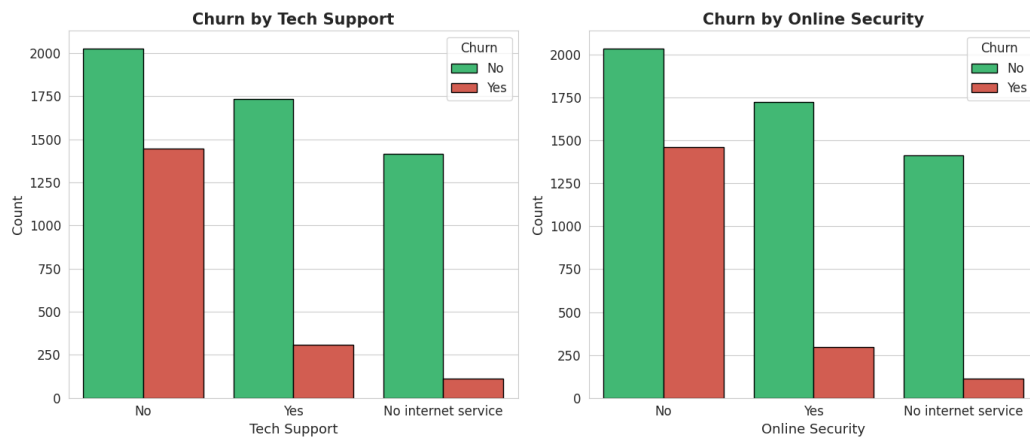


Figure 7: Churn by tech support and online security availability.

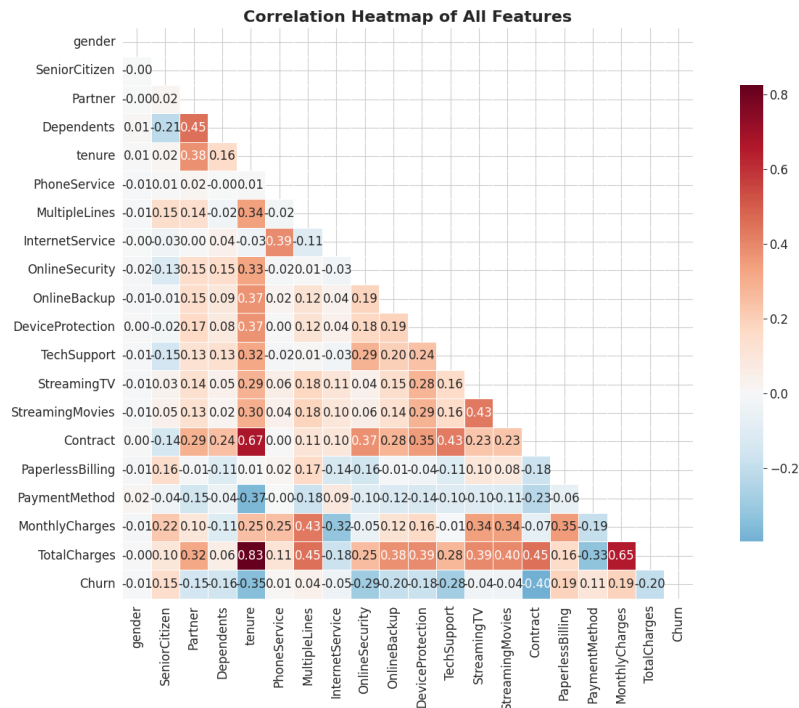


Figure 8: Correlation heatmap of all features against churn.

5.3 Statistical Hypothesis Testing Results

Table 3 summarizes the results of the three hypothesis tests conducted to validate the exploratory findings above.

Test	Comparison	Statistic	p-value	Conclusion
Chi-Square	Churn vs. Contract Type	$\chi^2 = 1184.60$	< 0.000001	Reject H_0 — churn is dependent on contract type
T-Test	Monthly Charges (churned vs. retained)	$T = 16.54$	< 0.000001	Reject H_0 — churned customers pay significantly more
Chi-Square	Churn vs. Internet Service	$\chi^2 = 732.31$	< 0.000001	Reject H_0 — churn is dependent on internet service type

All three tests are statistically significant at $p < 0.05$, confirming that contract type, monthly charges, and internet service type each have a genuine, non-random relationship with customer churn, directly supporting the patterns identified during exploratory analysis.

5.4 Machine Learning Model Comparison

Table 4 and Figure 9 summarize the performance of the four classification models on the held-out test set.

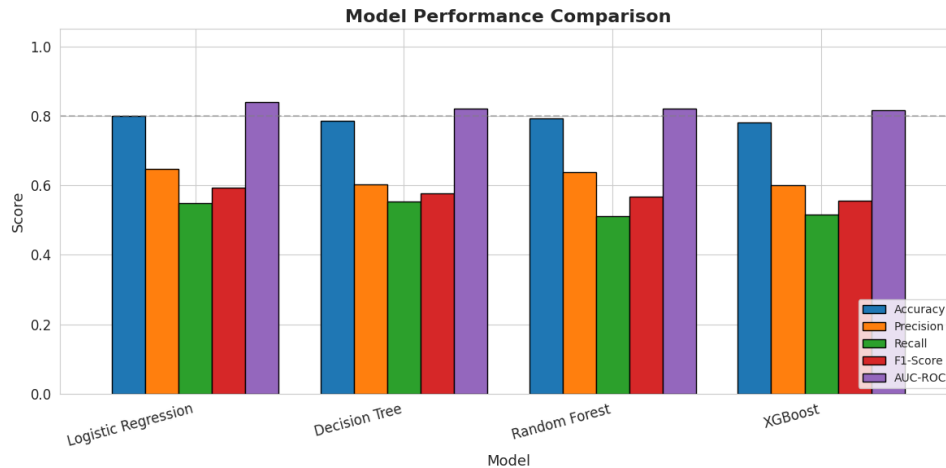


Figure 9: Model performance comparison across accuracy, precision, recall, F1-score, and AUC-ROC.

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Logistic Regression (best)	80.06%	64.67%	54.81%	59.33%	84.03%
Random Forest	79.35%	—	—	—	82.2%
Decision Tree	78.50%	—	—	—	82.2%
XGBoost	78.07%	—	—	—	81.6%

Logistic Regression achieved the best overall balance of performance, with the highest accuracy (80.06%) and the highest AUC-ROC (84.03%) among the four models. Although the ensemble models (Random Forest, XGBoost) are generally more flexible, they did not outperform the simpler linear baseline on this dataset, suggesting that the relationship between the selected features and churn is largely well captured by a linear decision boundary.

5.5 Best Model Evaluation: Confusion Matrix and ROC Curve

Figure 10 shows the confusion matrix for the best-performing model (Logistic Regression) on the test set. The model correctly identified 923 of 1,035 retained customers (89%) and 205 of 374 churned customers (55%), with 112 false positives and 169 false negatives. Figure 11 shows the ROC curve comparison across all four models; Logistic Regression achieves the highest AUC (0.840), indicating good discriminative ability between churners and non-churners, meaningfully better than a random classifier (AUC = 0.500).

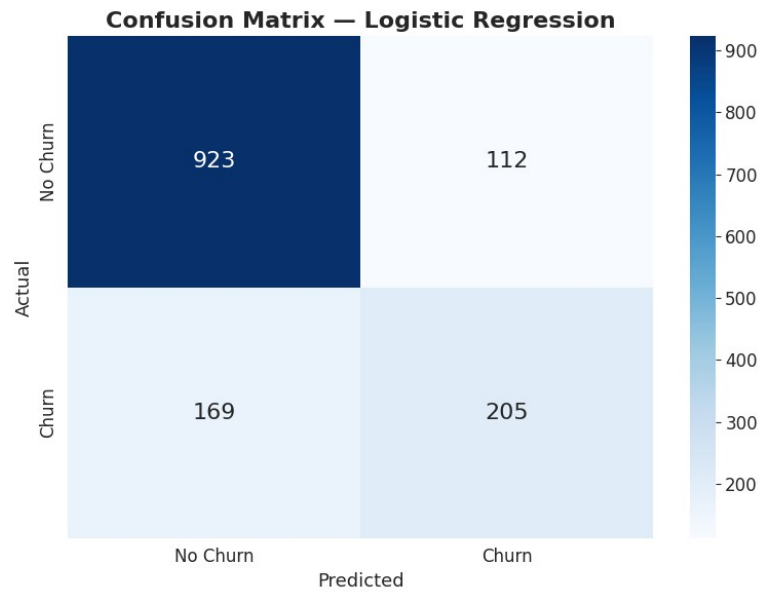


Figure 10: Confusion matrix — Logistic Regression (best model).

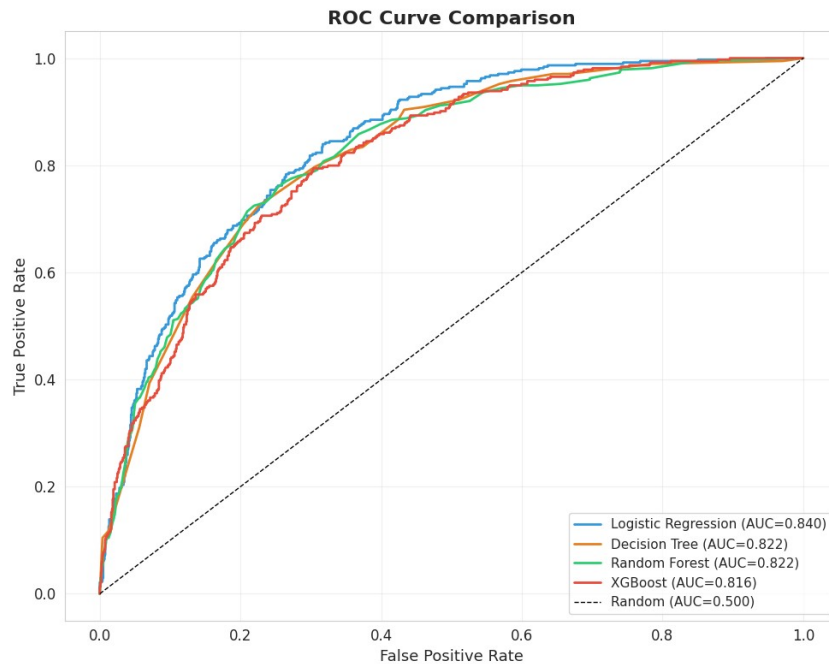


Figure 11: ROC curve comparison across all four models.

5.6 Feature Importance — Top Churn Drivers

Figure 12 ranks the top 15 features by importance score, derived from the trained models. Total Charges (0.187), Monthly Charges (0.179), and Tenure (0.154) emerged as the three strongest predictors of churn, followed by Contract Type (0.080) and Payment Method (0.050). This ranking is consistent with both the exploratory analysis (Section 5.2) and the hypothesis testing results (Section 5.3), reinforcing that billing level, customer tenure, and contract commitment are the dominant drivers of churn in this dataset.

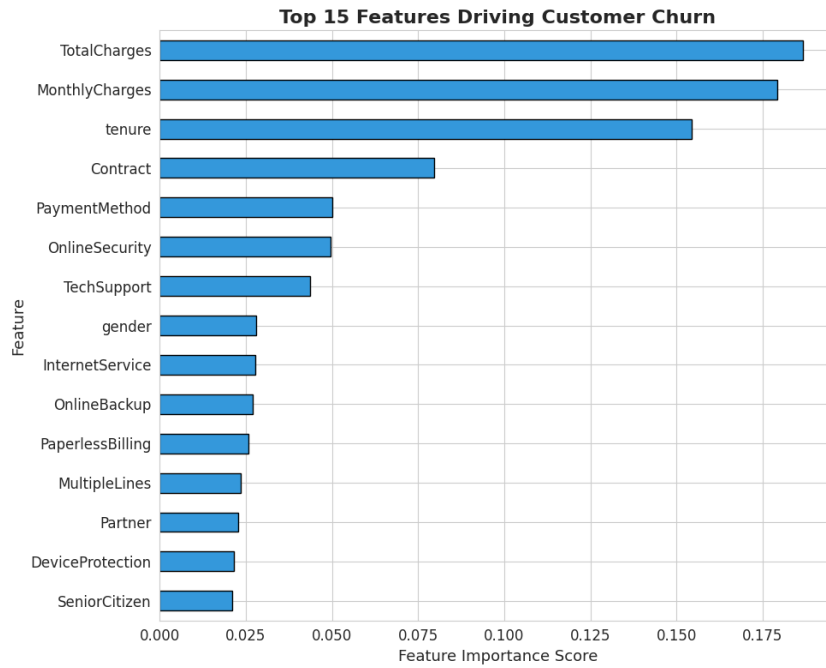


Figure 12: Top 15 features driving customer churn.

5.7 Summary of Research Question Findings

RQ	Evidence Used	Principal Finding
RQ1	Model comparison across accuracy, precision, recall, F1, AUC-ROC	Logistic Regression predicted churn with 80.06% accuracy and an AUC-ROC of 0.84, showing machine learning can classify churn risk with good discriminative accuracy using structured customer data.
RQ2	EDA, hypothesis testing, and feature importance analysis	Contract type, tenure, and monthly/total charges are the most strongly and significantly associated attributes with churn, each confirmed both statistically and via feature importance ranking.
RQ3	Risk drivers and model outputs combined with business recommendations	The identified drivers and model predictions provide a practical, actionable basis for targeted retention strategies, such as incentivizing longer contracts and prioritizing new customers for retention outreach.

6. Discussion

6.1 Interpretation of Findings

The results confirm that customer churn in this dataset is not randomly distributed but is concentrated among a specific, identifiable customer profile: short-tenure, month-to-month customers paying relatively high monthly charges without bundled support services. This profile emerged consistently across the exploratory analysis, the statistical hypothesis tests, and the feature importance rankings, giving strong triangulated confidence in the finding.

6.2 Comparative Model Performance

Contrary to the general expectation that ensemble methods such as Random Forest and XGBoost outperform simpler linear models, Logistic Regression achieved the best overall performance in this study. This suggests that the underlying relationship between the selected customer attributes and churn is largely linear and additive in nature, and that the added flexibility of tree-based ensembles was not necessary to capture it — an encouraging result from a deployment perspective, since Logistic Regression is also the most interpretable and computationally efficient of the four models tested.

6.3 Business Implications and Recommendations

The following recommendations follow directly from the statistical and model-based findings of this project:

11. Offer discounts or incentives for longer-term contracts to convert month-to-month customers, who churn at more than 15 times the rate of two-year contract holders.
12. Focus retention efforts on customers in their first 12 months of tenure, since new customers are consistently the highest-risk group.
13. Bundle tech support and online security as default or low-cost add-on services, given the large churn gap between customers with and without these services.
14. Review pricing and perceived value for high-monthly-charge customers, since higher bills are significantly associated with churn.
15. Encourage migration away from electronic check payments toward automatic payment methods, which are associated with substantially lower churn.

6.4 Comparison with Existing Literature

These findings are consistent with the broader churn prediction literature reviewed in Section 2, which similarly identifies contract structure, tenure, and billing level as dominant churn predictors (Verbeke et al., 2012), and which has previously shown that ensemble tree methods do not always outperform simpler baselines (Lemmens & Croux, 2006). This study extends that literature by combining formal statistical validation with a multi-model comparison on a real, structured telecom dataset.

6.5 Limitations

This study has several limitations. First, the dataset represents a single snapshot of customer data and does not capture how customer behavior or churn risk evolves over time. Second, the models were trained on a moderately sized dataset (7,043 records) and their generalization to other telecom markets or customer populations has not been tested. Third, feature importance rankings indicate association, not causation — the identified drivers should be treated as risk indicators for prioritizing retention outreach

rather than as proven causal mechanisms of churn. Finally, the recall of the best model (54.81%) indicates that a substantial share of customers who ultimately churn are not correctly flagged in advance, which should be considered when relying on the model for prioritization.

6.6 Future Research Directions

Future work could incorporate customer interaction data (e.g., support call frequency, complaint history) and temporal features that track how account attributes change over time, rather than relying on a single static snapshot. Threshold tuning and class-weighting techniques could also be explored to improve recall on the churn class, given the class imbalance observed in this dataset. Finally, periodic re-training and monitoring would be necessary before any operational deployment, to ensure the model remains accurate as customer behavior and market conditions evolve.

7. Conclusion

This project presented a complete, data-driven framework for understanding and predicting telecom customer churn, using a real dataset of 7,043 customer records. The framework combined data cleaning, exploratory data analysis, statistical hypothesis testing, and the comparative evaluation of four machine learning classification models.

The results successfully answered the research questions defined at the beginning of the study. The overall churn rate was found to be 26.5%, and three statistical hypothesis tests confirmed that contract type, monthly charges, and internet service type each have a significant, non-random relationship with churn ($p < 0.001$ in all cases). Among the four models tested, Logistic Regression achieved the best performance, with 80.06% accuracy and an AUC-ROC of 0.8403, and feature importance analysis identified Total Charges, Monthly Charges, and Tenure as the leading predictors of churn.

These findings were translated into concrete business recommendations, including incentivizing longer-term contracts, prioritizing retention outreach for new customers, and bundling support services as standard offerings. Although the study is limited by its reliance on a single dataset snapshot and by the moderate recall achieved on the churn class, the results demonstrate that a combination of statistically validated exploratory analysis and interpretable machine learning provides telecom companies with a practical, actionable foundation for proactive customer retention.

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Appendix: Representative Code Excerpts

This appendix reproduces representative code excerpts from Telecom_Customer_Churn_Analysis.ipynb to illustrate the implementation of key methodological steps described in Section 3.

A.1 Data Cleaning: TotalCharges Conversion

```
df['TotalCharges'] = df['TotalCharges'].replace(' ', np.nan)
df['TotalCharges'] = df['TotalCharges'].astype(float)
df['TotalCharges'].fillna(0, inplace=True)
```

A.2 Encoding and Scaling

```
le = LabelEncoder()
for col in categorical_cols:
    df_ml_encoded[col] = le.fit_transform(df_ml_encoded[col])

scaler = StandardScaler()
numerical_cols = ['tenure', 'MonthlyCharges', 'TotalCharges']
X[numerical_cols] = scaler.fit_transform(X[numerical_cols])

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

A.3 Model Training and Evaluation

```
models = {
    'Logistic Regression': LogisticRegression(max_iter=1000, random_state=42),
    'Decision Tree': DecisionTreeClassifier(random_state=42, max_depth=5),
    'Random Forest': RandomForestClassifier(n_estimators=100, random_state=42),
    'XGBoost': XGBClassifier(n_estimators=100, random_state=42,
eval_metric='logloss')
}

for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    y_prob = model.predict_proba(X_test)[:, 1]

    acc = accuracy_score(y_test, y_pred)
    prec = precision_score(y_test, y_pred)
    rec = recall_score(y_test, y_pred)
    f1 = f1_score(y_test, y_pred)
    auc = roc_auc_score(y_test, y_prob)
```

A.4 Statistical Hypothesis Testing

```
# Chi-Square test: Churn vs Contract Type
contingency = pd.crosstab(df['Contract'], df['Churn'])
chi2, p_value, dof, expected = chi2_contingency(contingency)

# T-Test: Monthly Charges (Churned vs Retained)
churned = df[df['Churn'] == 'Yes']['MonthlyCharges']
retained = df[df['Churn'] == 'No']['MonthlyCharges']
t_stat, p_value = ttest_ind(churned, retained)
```